Governing Trustworthy Agentic Healthcare AI: A Calibration-First Framework with Distributed Ledger Accountability

Michael O. Eniolade
University of the Cumberlands
meniolade20593@ucumberlands.edu

May 2026

Keywords: agentic AI; healthcare governance; probability calibration; distributed ledger technology; clinical decision support; trustworthy AI.

Abstract

Agentic artificial intelligence in healthcare is advancing faster than the governance frameworks designed to oversee it. These systems are autonomous, goal-directed, and capable of multi-step reasoning and sequential decision-making. They introduce accountability challenges that existing clinical AI standards were not built to address. Current standards, including TRIPOD+AI, DECIDE-AI, and the FDA guidance on AI/ML-based software as a medical device, were designed to evaluate static models at a point in deployment. They do not address pipeline-level reliability, compounding errors across agent decision steps, or the accountability vacuum created when autonomous systems act at speeds beyond human review.

This perspective proposes a five-layer trustworthiness governance framework for agentic healthcare AI that extends model validation standards to cover: (1) calibration integrity and uncertainty quantification at each agent decision point, (2) subgroup fairness auditing across demographic groups, (3) verifiable model provenance, (4) patient-controlled consent management for autonomous AI actions, and (5) scored deployment readiness with ongoing monitoring criteria. Each layer carries measurable thresholds. An Expected Calibration Error at or below 0.10, conformal prediction coverage at or above 0.90, and a subgroup ECE gap below 0.05 make the framework operationalizable rather than merely aspirational.

Blockchain and distributed ledger technologies (DLT) are not peripheral to this governance agenda. They provide the accountability infrastructure that makes trustworthiness evidence verifiable, persistent, and independently auditable. Immutable audit trails, smart contract-encoded ethical constraints, and decentralized consent registries address the epistemic accountability gap that traditional logging cannot fill. The framework connects directly to emerging empirical research on clinical AI calibration, conformal prediction, and deployment readiness evaluation, and is designed for empirical validation in subsequent research.

Plain Language Summary

What this paper is about. Healthcare AI is going "agentic." These systems make sequences of decisions without a human approving each step. An AI agent might monitor a patient's vital signs, detect a deterioration pattern, and automatically alert the care team or initiate escalation. These systems are already in clinical use.

The problem. The rules we use to keep healthcare AI safe were designed for tools that make one decision at a time. Those older rules do not address what happens when AI makes many consecutive decisions, how errors compound across steps, or who is accountable when an autonomous system causes harm.

What we propose. This paper proposes five requirements for any agentic AI system in healthcare: (1) probability predictions should be well-calibrated at every decision step, (2) the system should work equitably across patient groups, (3) training history and data sources should be traceable and verifiable, (4) patients should have control over what AI is permitted to do on their behalf, and (5) the system should pass a structured readiness assessment before deployment, with continuous monitoring thereafter.

Why blockchain helps. Blockchain creates permanent, tamper-proof digital records. This provides the foundation to make these five requirements enforceable and independently verifiable. Blockchain-based audit trails, smart contracts, and decentralized consent registries turn governance rules from policy documents into operational constraints built into the system itself.

Who this is for. Clinical informaticists, health system leaders, AI developers, and policymakers who need a practical governance framework for deploying agentic AI responsibly in healthcare.

1 Introduction

1.1 The Agentic Shift in Healthcare AI

Artificial intelligence in healthcare has crossed a threshold. For more than a decade, clinical AI research focused on a well-defined paradigm: train a model on historical data, validate it against a held-out test set, report discrimination metrics, and evaluate whether predictions exceed a performance threshold. The fundamental unit was a single inference: one patient, one input, one prediction. Clinical decision-support systems operating in this mode function as sophisticated calculators, responding to a clinician's query with a risk score or classification label.

This paradigm is being supplanted by a qualitatively different model of AI deployment. Agentic AI systems are autonomous, goal-directed software entities capable of multi-step reasoning, tool use, planning, and sequential action. They do not respond to a single prompt and stop. They pursue goals, decompose tasks, invoke external systems, retrieve information, make intermediate decisions, and produce outputs that often trigger further automated actions. Clinical examples span multiple care settings. Some systems autonomously monitor patient deterioration signals across data streams and initiate escalation protocols. Others iterate through laboratory and imaging results to refine differential diagnoses. Medication safety agents cross-check orders against patient records, interaction databases, and renal function values. Care coordination systems schedule and adjust clinical workflows without a clinician approving each step [1]. A 2025 systematic review of AI agents in clinical medicine documented twenty studies spanning discrete clinical tasks including diagnostic reasoning, medication management, and evidence retrieval, with agent systems outperforming their base model counterparts by a median of 53 percentage points in single-agent tool-calling studies, underscoring both the clinical promise and the governance urgency of agentic deployment [2].

1.2 Why This Changes the Governance Problem

The governance implications of this shift are not incremental. They are structural. When AI acts as a single-inference tool, accountability is straightforward: a clinician requests a prediction, evaluates it, and decides whether to act on it. The AI is a consultant whose opinion is accepted or rejected. When AI acts as an agent, this structure dissolves. The agent is no longer a consultant. It is a participant in care delivery. Its actions occur at speeds that preclude human review at each step. Its errors compound: an incorrect intermediate decision at step two affects all subsequent steps. Responsibility becomes diffuse across the AI system, the developers, the institution, and the clinicians nominally supervising the process.

1.3 The Governance Gap

Existing clinical AI governance frameworks were not designed for this environment. TRIPOD+AI [3] provides reporting standards for model development and validation but addresses models, not pipelines. DECIDE-AI [4] guides the evaluation of clinical decision-support interventions but does not distinguish between single-inference tools and multi-step agents. The FDA's AI/ML-based Software as a Medical Device (SaMD) framework [5] addresses pre-market validation and post-market surveillance but was conceived before agentic deployment became clinically common. The EU AI Act [6] imposes requirements for high-risk AI systems in healthcare but provides limited operational guidance for autonomous pipeline governance. None of these frameworks specifies how to measure, record, or verify trustworthiness at the level of individual agent decisions in a running clinical system.

This paper addresses that gap. We propose a five-layer trustworthiness governance framework specifically designed for agentic healthcare AI, and argue that distributed ledger technologies provide the accountability infrastructure necessary to make this framework operational and verifiable at scale.

2 What Makes Agentic Healthcare AI Different

Before governance frameworks can be designed, the distinctive properties of agentic AI that create governance challenges must be understood. Four properties differentiate agentic AI from traditional clinical prediction models.

2.1 Sequential Decision-Making and Error Compounding

Traditional clinical AI makes a single prediction given a single input. Agentic AI makes sequences of decisions, where each step's output feeds the next step's input. This creates an error compounding dynamic absent from single-inference systems: an overconfident probability estimate at step two, if not flagged and corrected, propagates into step four as if it were reliable. Calibration, the alignment between predicted probability and true event frequency, matters not only at the final output but at every intermediate decision that affects subsequent steps.

Current calibration evaluation methodology was developed for single-output models. A model's Expected Calibration Error (ECE) is computed over its test set predictions as a whole [7]. In an agentic pipeline, each decision node has its own calibration profile, and aggregate pipeline reliability depends on the product of individual node reliabilities. A pipeline composed of five individually well-calibrated decisions still produces systematically miscalibrated outputs if calibration error accrues in a consistent direction across nodes [8].

2.2 Opacity of Reasoning Chains

Single-inference models are analyzed using well-established interpretability methods: SHAP values decompose a prediction into feature contributions, calibration curves reveal probability reliability, and permutation importance identifies the features driving a classification. Agentic reasoning chains, particularly those involving large language model (LLM) components, iterative retrieval, or multi-agent orchestration, are substantially harder to audit. The intermediate states between initial input and final recommendation are partially or fully inaccessible, and their contribution to the final output often cannot be recovered after the fact.

This opacity creates an accountability deficit. When an agentic system produces a harmful recommendation, investigators are often unable to identify which reasoning step introduced the error, whether the error reflected a miscalibrated prediction, a biased training distribution, or an inappropriate action at a pipeline junction. Without a durable, auditable record of intermediate states, post-incident analysis becomes speculative.

2.3 Speed Beyond Human Review Cycles

A defining property of agentic AI is its capacity to act at speeds beyond human review. A triage agent processing 50 patient deterioration signals per hour cannot be meaningfully supervised by a clinician reviewing each action in real time. Meaningful human oversight requires mechanisms beyond real-time review: governance must be embedded in the system's design through pre-deployment constraints, hardcoded boundaries, and automatic escalation triggers, then verified through post-hoc audit rather than concurrent human review [9].

2.4 Diffuse Moral Responsibility

When a single prediction model is used by a clinician and a patient is harmed, responsibility analysis proceeds along a recognizable chain: the model developer, the deploying institution, and the clinician who acted on the recommendation. Agentic pipelines involving multiple AI components, orchestration layers, external data sources, and automated actions dissolve this chain. Responsibility becomes distributed across parties who each have partial control over some steps and no control over others [10]. This diffusion is not merely a legal problem. It creates practical barriers to learning from failures, holding actors accountable, and deterring unsafe deployment.

3 The Trustworthiness Gap in Current Governance Frameworks

3.1 What Existing Frameworks Address

Several frameworks have emerged to govern clinical AI quality. TRIPOD+AI [3] extends the TRIPOD reporting guideline to include AI-specific transparency requirements: model architecture, training data provenance, calibration reporting, and external validation. DECIDE-AI [4] provides guidance for the evaluation of AI-based clinical decision-support interventions, emphasizing clinical utility alongside statistical performance. The FDA SaMD framework [5] establishes a risk-based regulatory pathway for AI-enabled medical devices, including requirements for pre-market validation and post-market surveillance. The EU AI Act [6] classifies healthcare AI as high-risk and imposes requirements for transparency, documentation, human oversight, and robustness.

3.2 What They Miss

Despite their importance, these frameworks share a common limitation: they were designed for model-level evaluation at a point in time, not for ongoing pipeline-level governance of autonomous systems.

Calibration is underspecified. TRIPOD+AI recommends calibration reporting but provides no threshold standards. A 2024 systematic review and meta-analysis of prediction models for incident chronic kidney disease derived or validated in community-based electronic health records found calibration reporting so deficient that pooled meta-analysis was not possible. Risk of bias was high in 64% of included models, and no clinical utility analyses or impact studies were identified for any model [11]. The implication for agentic AI governance is direct: existing frameworks cannot ensure the probability estimates produced by agent decision nodes are reliable enough to support downstream automated actions.

Uncertainty is absent. Fewer than 4% of clinical AI studies report any uncertainty measure alongside their predictions [12]. For single-inference tools, this is a quality gap. For agentic systems where uncertain predictions propagate into subsequent automated actions, it is a safety gap. If an agent does not know, and cannot communicate, that its probability estimate at step three carries high uncertainty, it cannot appropriately defer to human review or widen its decision threshold.

Fairness is aggregate, not pipeline-level. Where fairness evaluation exists in clinical AI governance, it typically measures whether subgroup performance gaps exist at the level of a model's test set. It does not evaluate whether bias accrues or compounds across agent pipeline steps, or whether demographic disparities in agent recommendations emerge from the interaction of multiple individually unbiased components [13].

Audit trails are informal and mutable. No current governance framework specifies what constitutes an adequate audit record for an AI agent's actions. Server logs, model outputs, and database entries are institution-specific, mutable, and not independently verifiable. In a regulatory dispute or adverse event investigation, the absence of immutable, standardized audit records makes accountability analysis unreliable.

Consent frameworks predate autonomy. Existing informed consent mechanisms in healthcare AI were designed around patient consent to data use and model deployment. They were not built for patient consent to autonomous AI action on their behalf at each step of a clinical process.

As agentic AI increasingly acts without direct human mediation, the concept of meaningful consent requires rethinking [13].

4 A Five-Layer Trustworthiness Governance Framework

We propose a five-layer framework that addresses the gaps identified above. Each layer carries measurable criteria and applies both pre-deployment and as a continuous monitoring protocol for running agentic systems.

4.1 Layer 1: Reliability Evaluation

Reliability evaluation governs whether the probability outputs produced by each agent decision node are trustworthy enough to support downstream automated action.

Calibration assessment requires computing the Expected Calibration Error (ECE) for each decision node using empirical frequency analysis across probability bins [7]. We propose ECE ≤ 0.10 as a minimum deployment threshold, consistent with emerging clinical AI quality standards. The Brier Score and Brier Skill Score provide complementary summary measures. Guo et al. [7] demonstrate how both metrics reveal systematic overconfidence in deep learning models. Critically, calibration must be evaluated not only on the original training distribution but on the distribution of inputs the deployed agent actually encounters, including the heterogeneous, missing-data-rich inputs characteristic of real clinical environments.

Uncertainty quantification using split conformal prediction [8] provides distribution-free coverage guarantees at each decision node. Conformal prediction produces a prediction set rather than a single probability estimate. This set is guaranteed to contain the true label with probability $1 - \alpha$, where α is the chosen miscoverage rate. We propose a coverage guarantee of ≥ 0.90 for clinical agentic AI, ensuring the agent communicates not just its best prediction but the range of outcomes it cannot rule out. When prediction sets include multiple labels, indicating ambiguous predictions, the agent should defer to human review rather than proceed autonomously.

External validation is required before deployment: calibration metrics should be recomputed on a prospectively collected or geographically distinct external dataset to measure calibration drift, the degradation of probability reliability when the model encounters a new clinical environment [11].

4.2 Layer 2: Fairness Auditing

Fairness auditing governs whether agent reliability is equitably distributed across patient subgroups.

Calibration should be stratified by age group (e.g., below 65 vs. 65 and older), sex, race/ethnicity where available, and clinically relevant comorbidity groups (e.g., diabetes status, chronic kidney disease stage). We propose a maximum allowable ECE gap of 0.05 between any two demographic subgroups as a deployment gate. Where subgroup calibration gaps exceed this threshold, the deploying institution should either perform targeted recalibration on underrepresented groups or restrict deployment to populations where calibration has been demonstrated.

False negative disparity deserves particular attention in healthcare AI, where missed diagnoses disproportionately harm already-disadvantaged groups [13]. An agent well-calibrated overall but systematically underestimating risk for elderly patients or patients from racial minority groups amplifies existing healthcare inequities rather than reducing them.

4.3 Layer 3: Provenance Verification

Provenance verification governs whether the lineage of an agent's models and training data is independently verifiable.

Model versioning requires each deployed agent decision node to carry a verifiable identifier, a cryptographic hash of the model weights and architecture, recorded at deployment. Any change to the model, including recalibration, fine-tuning, or architecture modification, must generate a new identifier and a new deployment record.

Training data documentation requires that the characteristics of training datasets, including size, demographic composition, data source, collection period, and preprocessing steps, be documented and verifiable. This documentation must allow an independent reviewer to assess whether the training distribution is representative of the deployment environment.

Deployment history records when each model version was deployed, to which clinical contexts, and under what oversight conditions. This record enables post-incident analysis: when a patient is harmed, investigators identify exactly which model version was running, what it was trained on, and whether its deployment conditions matched its validation conditions.

4.4 Layer 4: Consent Management

Consent management governs the scope of autonomous AI action to which a patient has agreed.

Current consent frameworks in healthcare AI address consent to data use and model deployment at the institutional level. Agentic AI requires more granular consent: a patient should be able to specify whether they consent to autonomous AI actions affecting their care plan, whether they require human review before AI-initiated actions take effect, and whether they retain the right to override an AI agent's recommendation at any step.

This requires a consent architecture capable of expressing, storing, and enforcing these preferences, and updating them as the patient's care context changes. Static institutional consent forms signed at admission are insufficient for an environment where AI agents act on a patient's behalf across multiple care episodes, data systems, and clinical teams.

4.5 Layer 5: Deployment Readiness and Ongoing Monitoring

Deployment readiness scoring integrates the preceding four layers into a structured pre-deployment gate and a continuous post-deployment monitoring protocol.

Pre-deployment, each agent is assessed against a deployment readiness checklist:

1. Discrimination adequacy (AUROC ≥ 0.85 on external validation)
2. Calibration adequacy (ECE ≤ 0.10 on external validation)
3. Calibration stability (calibration drift ≤ 0.05 across internal and external datasets)
4. Uncertainty coverage (conformal coverage ≥ 0.90 on external validation)

5. Coverage stability (coverage drift ≤ 0.05 across datasets)
6. Prediction interpretability (singleton rate ≥ 0.70, the proportion of instances receiving an unambiguous prediction)
7. Subgroup calibration equity (ECE gap across subgroups ≤ 0.05)
8. Transparency and reproducibility (full code, data pipeline, and model documentation publicly or institutionally accessible) [14]

Post-deployment, ongoing monitoring tracks calibration drift over time using sequential ECE monitoring with statistical process control alerts. A statistically significant increase in ECE above the deployment threshold triggers a mandatory review cycle and, if unresolved, mandatory decommissioning.

5 Distributed Ledger Technologies as Accountability Infrastructure

5.1 Why Traditional Logging Is Insufficient

The trustworthiness framework described above generates a substantial body of evidence: calibration metrics, fairness audits, provenance records, consent documents, deployment readiness scores, and ongoing monitoring data. For this evidence to support accountability in regulatory review, adverse event investigation, or litigation, it must be durable, verifiable, and independent of any single institution's control.

Traditional server logs and database records fail on all three criteria. They are mutable: records can be altered by administrators with access. They are siloed: records from different components of a multi-institutional agentic pipeline are stored in incompatible systems without a common reference. They are not independently verifiable: an institution claiming its AI agent behaved correctly at a given time cannot prove its own internal records have not been modified.

These limitations are not merely theoretical. In healthcare environments where AI agents act across institutional boundaries, for example a diagnostic agent querying a regional radiology system or a triage agent integrating data from multiple hospitals, no single institution's logging architecture provides an authoritative audit record of the full pipeline.

5.2 Blockchain-Based Audit Trails

Blockchain and distributed ledger technologies address these limitations through properties architecturally suited to the accountability requirements of agentic healthcare AI. As blockchain and AI converge as complementary pillars of digital health transformation, their integration for governance and accountability has emerged as a priority challenge for healthcare systems globally [15].

Immutability means that once a transaction is written to the ledger, it cannot be altered without consensus from the network, making post-hoc modification of audit records computationally and organizationally prohibitive. For agentic AI governance, each significant agent decision (a risk score above a clinical threshold, an automated order, a care plan modification) is recorded as an on-chain transaction, creating a tamper-resistant record of what the agent did, when, on what inputs, and with what output.

Verifiability means any authorized party independently confirms that a record has not been modified since it was written. A regulatory auditor, a patient's legal representative, or a clinical ethics committee verifies the audit trail of an AI agent's actions without depending on the deploying institution's attestation.

Decentralization means the audit record is not controlled by any single party. In a multi-institutional agentic pipeline, a shared distributed ledger allows each participant to contribute to and verify the record without any single party having unilateral control over what the record says [16].

Smart contracts enable the encoding of governance rules as executable code running on the ledger. Rather than relying on institutional policy, a smart contract enforces governance rules automatically: if an agent's conformal prediction set contains multiple labels, the contract blocks autonomous action and generates a human escalation event recorded on-chain. Ullah et al. demonstrate this approach in the EHR domain, implementing purpose-based access control policies through smart contracts that establish comprehensive, immutable audit trails recording transaction details, access purposes, digital signatures, and timestamps, providing accountability for all data access events while eliminating dependence on trusted intermediaries [16]. Governance rules become operational constraints with independently verifiable enforcement, not aspirational policies.

5.3 Decentralized Consent Management

Patient consent for autonomous AI action is particularly well-suited to blockchain-based management. A consent record written to a distributed ledger is patient-controlled, institution-independent, and auditable. A patient who consents to AI-assisted triage but not AI-initiated medication changes records this granular preference in a consent entry that any component of the agentic pipeline checks before acting.

Smart contracts enforce consent boundaries in real time: when an AI agent proposes an action, the contract checks whether that action class falls within the patient's recorded consent scope and blocks it if not, generating a logged exception for human review. This architecture gives consent teeth. Consent is not simply a document signed at admission but a live constraint on system behavior.

5.4 Verifiable Model Provenance

The provenance requirements described in Layer 3 are directly implementable using on-chain model registries. When a new model version is deployed, its cryptographic hash, computed from the model weights, architecture specification, and training data characteristics, is written to the ledger. Any subsequent query to the agent verifies, in real time, that the model responding to the query is the registered, validated version and not a modified derivative.

This is particularly important in environments where models are periodically updated: a post-deployment recalibration that improves performance on a new patient population sometimes degrades performance on a previously studied subgroup. An on-chain model registry makes every version change traceable, reversible, and independently verifiable. Liu and Hu's systematic review of blockchain-AI integration in healthcare, spanning 1,221 publications from 2015 to 2025, identifies model provenance, traceability, and smart contract-enforced data governance as foundational requirements for building a trustworthy and intelligent healthcare ecosystem, with blockchain providing the immutability and auditability that AI-generated clinical outputs require [17].

5.5 Federated Learning for Privacy-Preserving Distributed Governance

Governance at scale, across multiple hospitals, health systems, and jurisdictions, requires access to diverse patient populations for calibration validation and fairness auditing. Federated learning enables models to be trained and calibrated across distributed datasets without centralizing sensitive patient data, with distributed ledger technologies providing the coordination, verification, and audit infrastructure for the federated process [18].

Each participating institution trains on its local data, contributes gradient updates or model statistics to the federated process, and receives in return a globally calibrated model whose performance has been verified across the federation's full population diversity. The ledger records each institution's participation, the aggregation protocol used, and the resulting model provenance, providing accountability for the full federated governance process.

6 Implementation Challenges

The framework proposed here is architecturally sound, but its implementation faces challenges that a research agenda must address honestly.

6.1 Latency and Throughput

Writing audit transactions to a distributed ledger introduces latency that is often incompatible with real-time clinical decision support. A triage agent writing an on-chain transaction for each decision step faces throughput constraints that delay time-sensitive interventions. Architectural solutions including off-chain computation with on-chain commitment, periodic batch anchoring of audit summaries, and layer-2 scaling protocols help mitigate this, but they require careful design to preserve the immutability properties that motivate on-chain governance in the first place.

6.2 Interoperability with Existing Health Information Systems

Healthcare institutions operate across a heterogeneous collection of electronic health record systems, laboratory information systems, and clinical data warehouses that predate blockchain-based architectures. Integrating on-chain governance with existing health information infrastructure requires FHIR-compliant data exchange, HL7 interface standards, and institutional commitment to adopting new auditing protocols. On-chain governance cannot be imposed on existing systems without substantial integration engineering.

6.3 GDPR and the Right to Erasure

Blockchain's immutability creates a direct tension with GDPR's right to erasure. A patient who invokes the right to have their data deleted cannot have that data removed from an immutable ledger. One architectural approach stores only cryptographic commitments on-chain while keeping personal data off-chain. Erasure is implemented by deleting the off-chain data and rendering the commitment unresolvable, preserving both the audit record's integrity and the practical effect of erasure. Bai et al. demonstrate this hybrid model in practice for smart healthcare systems, using IPFS for off-chain storage of personally identifiable and protected health information while maintaining only encrypted references on the blockchain ledger, implementing GDPR Article 17 compliance by enabling deletion from IPFS storage without compromising the integrity of the on-chain audit record [19]. The broader legal adequacy of these hybrid approaches across different regulatory jurisdictions remains an open question.

6.4 Institutional Adoption and Governance

Distributed ledger governance requires consortium formation among participating institutions: agreement on ledger architecture, consensus mechanisms, access control, and governance of the governance layer itself. Healthcare institutions accustomed to managing their own data infrastructure often resist participation in a shared ledger they do not unilaterally control. Building the institutional trust and regulatory legitimacy for cross-institutional blockchain-based AI governance is a sociotechnical challenge that technical architecture alone cannot solve.

6.5 Clinician Trust Calibration

The governance framework described here is designed primarily around system-level accountability. Agentic AI governance also has a human dimension: clinicians must be neither over-reliant on AI agent recommendations nor systematically distrustful of them. Calibration, in this context, refers not only to probability reliability but to the alignment between a clinician's confidence in an AI recommendation and the AI's actual reliability. Governance frameworks that improve system trustworthiness must also invest in clinician education about when and how to appropriately weight AI recommendations.

7 Research Agenda

The framework proposed here is a conceptual contribution that requires empirical validation. Five research priorities follow from it.

Priority 1: Empirical calibration evaluation of deployed clinical AI. The framework's reliability layer requires that calibration be measured and documented for each agent decision node. Large-scale empirical studies measuring ECE, Brier scores, and conformal coverage rates across deployed clinical AI systems, including agentic pipelines, would establish the current state of the field and identify where governance intervention is most urgently needed. The finding that calibration reporting was insufficient for meta-analysis even in a 2024 systematic review of CKD prediction models [11] illustrates how far current practice falls from the governance standard the framework requires.

Priority 2: Smart contract frameworks for clinical AI ethical constraints. The formal specification of governance rules in smart contract code, particularly escalation triggers, consent enforcement, and calibration drift alerts, requires the development of healthcare-specific smart contract templates validated against clinical governance requirements. Piloting these frameworks in simulated clinical environments would provide evidence of operational feasibility.

Priority 3: Cross-institutional federated governance pilots. The federated learning and distributed governance architecture described in Section 5.5 requires multi-institutional pilots to evaluate feasibility, latency, interoperability, and governance dynamics. Such pilots would also generate the diverse external validation datasets needed to properly evaluate calibration stability and subgroup fairness across institutional settings.

Priority 4: Patient-facing uncertainty communication standards. If conformal prediction is to be used to communicate uncertainty to patients and clinicians in a way that supports informed consent and meaningful oversight, standards are needed for how uncertainty should be displayed, interpreted, and acted upon in clinical interfaces. Human factors research on uncertainty communication in clinical decision support would bridge the technical and human dimensions of the governance framework.

Priority 5: Prospective evaluation of DLT audit trails in adverse event investigation. The value of blockchain-based audit trails ultimately depends on evidence from actual adverse event investigations. Prospective studies comparing the completeness, accuracy, and usefulness of on-chain versus conventional audit records in clinical AI incident reviews would provide the evidence base needed to justify the significant investment required for DLT adoption in healthcare AI governance.

8 Conclusion

Agentic AI in healthcare is not a future prospect. It is a present reality. Systems that autonomously monitor, reason, and act on behalf of patients are already in clinical use, and their capabilities are expanding rapidly. The governance frameworks currently available to healthcare institutions, regulators, and patients were designed for a simpler world: a model produces a prediction, a clinician evaluates it, a patient receives care. AI agents now make sequential decisions at machine speed. Those frameworks leave critical accountability gaps unfilled.

The five-layer trustworthiness governance framework proposed here addresses those gaps with measurable, operationalizable criteria. It integrates reliability evaluation, fairness auditing, provenance verification, consent management, and deployment readiness scoring. Distributed ledger technologies provide the accountability infrastructure that makes this framework verifiable: immutable audit trails, smart contract-enforced constraints, decentralized consent registries, and on-chain model provenance records transform governance from aspirational policy into operational architecture.

Trustworthiness in agentic healthcare AI cannot be certified once at deployment and assumed thereafter. It must be earned continuously, measured rigorously, and verified independently. The framework presented here offers one path toward that goal, alongside a research agenda for the work required to validate it.

Acknowledgments

The author acknowledges the growing body of clinical AI governance scholarship that informed this perspective, particularly the TRIPOD+AI, DECIDE-AI, and emerging conformal prediction frameworks for clinical uncertainty quantification. Portions of this manuscript were drafted with the assistance of Claude (Anthropic), an AI language model, used for structural drafting, editorial refinement, and reference formatting. The author takes full responsibility for all content, interpretations, and conclusions.

Financial Disclosures

Funding: None.

Conflicts of interest: The author declares no financial or non-financial conflicts of interest relevant to the content of this manuscript.

Contributors

Michael O. Eniolade contributed to conceptualization, framework development, formal analysis, writing – original draft, and writing – review and editing.

Data Availability Statement

No new data were generated or analyzed in this study. This is a conceptual perspective paper. All supporting evidence is drawn from published literature cited in the references.

References

1. Rajpurkar P, Chen E, Banerjee O, Topol EJ. AI in health and medicine. Nat Med. 2022;28(1):31–38. https://doi.org/10.1038/s41591-021-01614-0

2. Gorenshtein A, Omar M, Glicksberg BS, Nadkarni GN, Klang E. AI Agents in Clinical Medicine: A Systematic Review. medRxiv. 2025;2025.08.22.25334232. https://doi.org/10.1101/2025.08.22.25334232

3. Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. https://doi.org/10.1136/bmj-2023-078378

4. Vasey B, Nagendran M, Campbell B, et al. Reporting guideline for the early stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. BMJ. 2022;377:e070904. https://doi.org/10.1136/bmj-2022-070904

5. U.S. Food and Drug Administration. Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan. Silver Spring, MD: FDA; 2021.

6. European Parliament and Council. Regulation (EU) 2024/1689 of the European Parliament and of the Council — Artificial Intelligence Act. Official Journal of the European Union. 2024.

7. Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. Proc 34th Int Conf Mach Learn. 2017;70:1321–1330.

8. Angelopoulos AN, Bates S. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. Found Trends Mach Learn. 2023;16(4):494–591. https://doi.org/10.1561/2200000101

9. Choudhury A, Asan O. Role of artificial intelligence in patient safety outcomes: systematic literature review. JMIR Med Inform. 2020;8(7):e18599. https://doi.org/10.2196/18599

10. Morley J, Cowls J, Taddeo M, Floridi L. Ethical guidelines for AI in public service. Gov Inf Q. 2021;38(3):101577. https://doi.org/10.1016/j.giq.2021.101577

11. Haris M, Raveendra K, Travlos CK, Lewington A, Wu J, Shuweidhi F, Nadarajah R, Gale CP. Prediction of incident chronic kidney disease in community-based electronic health records: a systematic review and meta-analysis. Clin Kidney J. 2024;17(5):sfae098. https://doi.org/10.1093/ckj/sfae098

12. Campagner A, Ciucci D, Cabitza F. Modeling the unknown: on the role of uncertainty in machine learning for healthcare. Comput Methods Programs Biomed. 2025. https://doi.org/10.1016/j.cmpb.2024.108576

13. Mittelstadt B, Allo P, Taddeo M, Wachter S, Floridi L. The ethics of algorithms: mapping the debate. Big Data Soc. 2016;3(2):2053951716679679. https://doi.org/10.1177/2053951716679679

14. Collins GS, Dhiman P, Ma J, et al. Evaluation of clinical prediction models (part 1): from development to external validation. BMJ. 2024;384:e074819. https://doi.org/10.1136/bmj-2023-074819

15. Dershem M, Riley III J, Venkataraman M, Nasr J, Hussain AS. 2026 Healthcare Predictions: AI, Blockchain, and the Rise of Decentralized Innovation. BHTY. 2025;8(3). https://doi.org/10.30953/bhty.v8.475

16. Ullah F, He J, Zhu N, Wajahat A, Nazir A, Qureshi S, Pathan MS, Dev S. Blockchain-enabled EHR access auditing: Enhancing healthcare data security. Heliyon. 2024;10(16):e34407. https://doi.org/10.1016/j.heliyon.2024.e34407

17. Liu J, Hu X. Blockchain meets AI in healthcare: a review of convergent technologies for digital health transformation. Front Blockchain. 2026;9:1766092. https://doi.org/10.3389/fbloc.2026.1766092

18. Rieke N, Hancox J, Li W, et al. The future of digital health with federated learning. npj Digit Med. 2020;3(1):119. https://doi.org/10.1038/s41746-020-00323-1

19. Bai P, Kumar S, Kumar K, Kaiwartya O, Mahmud M, Lloret J. GDPR Compliant Data Storage and Sharing in Smart Healthcare System: A Blockchain-Based Solution. Electronics. 2022;11(20):3311. https://doi.org/10.3390/electronics11203311